TCSS 321 – Discrete Structures

Formal Proofs

Proofs

Definition: A proof is a valid and sound argument that establishes the truth of a proposition (conclusion) from a set of assumed propositions (premises).

Definition 1: An argument is valid if and only if the conclusion is true when the premises are true.

Definition 2: An argument is sound if and only if the premises are true.

Theorem: If an argument is both valid and sound then the conclusion is true.

Proof: Assume an argument is both valid and sound.

By the definition of valid if the premises are true then the conclusion is true.

By the definition of sound the premises are true.

Therefore the conclusion is true. ■

- This is an informal proof.

Premises

Definition: A premise is a proposition that is assumed to be true prior to the proof.

- A premise may be an axiom, theorem, or hypothesis.

Definition: An axiom is a proposition that is assumed to be true without proof.

- Definitions are assumed without proof.

Definition: A theorem is a proposition that has already been proven true (or we are about to prove true).

- We use theorems in our proofs in order to prove other theorems.

Definition: A hypothesis is a proposition that is assumed true for the sake of the argument.

- If the hypothesis is not true then the proof is not sound and establishes nothing.
- If the hypothesis is true then the conclusion is true too (assuming a valid proof).

Direct Proofs

Definition: A direct proof establishes a conclusion assuming some premises.

- Most proofs are direct proofs.

- Assume $P = \{p_1, p_2, \ldots, p_n\}$ is the set of premises.
- Assume c is the conclusion.
- Then every direct proof is trying to prove that $(p_1 \wedge p_2 \wedge \cdots \wedge p_n) \rightarrow c$.

- In a direct proof we always assume that the premises $p_1 \wedge p_2 \wedge \cdots \wedge p_n$ are true and argue that the consequent c is true.

- A direct proof must always use valid rules of inference.

Definitions

- A definition always establishes a biconditional (if an only if) relation between a term t and a property $P(x)$.
  - $P(x)$ means that "x has property P".
  - P is commonly called a predicate.

Definition: An x is a t if and only if $P(x)$.

Definition: A number x is an even natural number if and only if $x = 2k$ where $k \in \mathbb{N}$.
  - Here the term t is "an even natural number" and the property $P(x)$ is "$x = 2k$ where $k \in \mathbb{N}$".

- As a biconditional definitions establish two conditionals.
  - If x is a t then $P(x)$.
  - If $P(x)$ then x is a t.

Formal Proofs

Let v be the proposition "the argument is valid" and s be the proposition "the argument is sound".

Let p be the proposition "the premises are true" and c be the proposition "the conclusion is true".

Definition 1: An argument is valid if and only if the conclusion is true when the premises are true.

$$v \leftrightarrow (p \rightarrow c)$$

Definition 2: An argument is sound if and only if the premises are true.

$$s \leftrightarrow p$$

Theorem: If an argument is both valid and sound then the conclusion is true.

$$(v \wedge s) \rightarrow c$$

Proof: Assume $(v \wedge s)$. *(hypothesis)*

Therefore v and therefore s. *(conjunctive simplification)*

By Definition 1 $(p \rightarrow c)$. *(modus ponens)*

By Definition 2 p. *(modus ponens)*

Therefore c. *(modus ponens)* ■

- This is a formal proof.

Rules of Inference

Formal Name	Rule	Tautology
Modus ponens	$p \to q$ p $\therefore q$	$\big((p \to q) \land p\big) \to q$
Modus tollens	$p \to q$ $\neg q$ $\therefore \neg p$	$\big((p \to q) \land \neg q\big) \to \neg p$
Hypothetical syllogism	$p \to q$ $q \to r$ $\therefore p \to r$	$\big((p \to q) \land (q \to r)\big) \to (p \to r)$
Disjunctive syllogism	$p \lor q$ $\neg p$ $\therefore q$	$\big((p \lor q) \land \neg p\big) \to q$

Formal Name	Rule	Tautology
Addition	p $\therefore p \lor q$	$p \to (p \lor q)$
(Conjunctive) Simplification	$p \land q$ $\therefore p$	$(p \land q) \to p$
Conjunction	p q $\therefore p \land q$	$(p \land q) \to (p \land q)$
Resolution	$p \lor q$ $\neg p \lor r$ $\therefore q \lor r$	$\big((p \lor q) \land (\neg p \lor r)\big) \to q \lor r$

Theorems

- There are many names for propositions depending on what we know about them or how we plan to use them in our proofs.

Definition: A theorem is a proposition that has been proven true.

- A theorem is also what we name a proposition that we are about to prove.

Definition: A conjecture or hypothesis is a proposition that has not yet been proven true.

- We will often assume a hypothesis is true in order to prove consequences of the hypothesis.

Definition: A corollary is a theorem that follows immediately from another theorem.

- We usually omit a proof of corollaries as they are trivial.

Definition: A lemma is a less important theorem proven to establish a second more important theorem.

- A proof of a proposition like $p \leftrightarrow q$ usually proceeds in two lemmas first proving $p \rightarrow q$ and then proving $q \rightarrow p$.

Proofs with sets

- Let S and T be sets.

Definition 3: $A \subseteq B$, if and only if, if $a \in A$ then $a \in B$.
Definition 4: $A = B$, if and only if, $A \subseteq B$ and $A \subseteq B$.

Theorem: If $S \subseteq T$, then $S = S \cap T$.
Proof: Assume $S \subseteq T$. So if $x \in S$ then $x \in T$.
Lemma 1: $S \subseteq S \cap T$.
Assume $x \in S$. Then $x \in T$. So $x \in S \wedge x \in T$. So $x \in S \cap T$. So if $x \in S$ then $x \in S \cap T$. So $S \subseteq S \cap T$.
Lemma 2: $S \cap T \subseteq S$.
Assume $x \in S \cap T$. Then $x \in S \wedge x \in T$. So $x \in S$. So if $x \in S \cap T$ then $x \in S$. So $S \cap T \subseteq S$.
So $S = S \cap T$. ■

Proofs with Sets

Let s be the proposition "$x \in S$" and let t be the proposition "$x \in T$" and let b be the proposition "$x \in S \cap T$".

Let u be the proposition "$S \subseteq T$" and let e be the proposition "$S = S \cap T$".

Let e_1 be the proposition "$S \subseteq S \cap T$" and let e_2 be the proposition "$S \cap T \subseteq S$".

Definition 3: $A \subseteq B$, if and only if, if $a \in A$ then $a \in B$.

$$u \leftrightarrow (s \rightarrow t)$$

Definition 4: $A = B$, if and only if, $A \subseteq B$ and $A \subseteq B$.

$$e \leftrightarrow (e_1 \wedge e_2)$$

Theorem 2: If $S \subseteq T$, then $S = S \cap T$.

$$u \rightarrow e$$

Proof: Assume u. (hypothesis) So if s then t. (by Definition 3)

Lemma 1: e_1. Assume s. (hypothesis) Then t. (Definition 3 and modus ponens) So $s \wedge t$. (conjunction) So b. (Definition of $\cap$) So $s \rightarrow b$. (Definition of $\rightarrow$) So e_1. (Definition 3)

Lemma 2: e_2. Assume b. (hypothesis) Then $s \wedge t$. (Definition of $\cap$) So s. (simplification) So $b \rightarrow s$. (Definition of $\rightarrow$) So e_2. (Definition 3)

So e. (Definition 4) ■

Formal and Informal Proofs

Definition: A formal proof states each step of the argument using propositional logic and valid rules of inference.

Definition: An informal proof might skip steps that appear obvious or trivial and may omit formalization in propositional logic (but must still only carry out valid steps).

- An informal proof is usually stated in a mix of natural language and formal mathematical notation.

- Both previous proofs were stated both formally and informally.
  - It is rare to state a proof formally because they tend to be long and tedious.

- We will use formal proofs until we get used to them and then we will use only informal proofs.

Proofs with Set of Numbers

- Can we prove this theorem about the even numbers $\mathbb{E}$?

Theorem: If $n, m \in \mathbb{E}$ then $nm \in \mathbb{E}$.

Proof: Assume $n, m \in \mathbb{E}$.

So $n = 2k$ and $m = 2l$ where $k, l \in \mathbb{N}$.

Then $nm = (2k)(2l) = 2(2kl)$.

So $nm \in \mathbb{E}$. ■

- This proof demonstrates the closure of $\mathbb{E}$ under the operation $\cdot$.
  - It relies on the definition of $\mathbb{E}$, the associative and commutative laws of arithmetic and the closure of $\mathbb{N}$ under the operation $\cdot$.

Proofs with Set of Numbers

- Can we formalize this proof?

Let p be the proposition "$n \in \mathbb{E}$" and q be the proposition "$m \in \mathbb{E}$".
Let r be the proposition "$n = 2k$" and s be the proposition "$m = 2l$" where $k, l \in \mathbb{N}$.
Let t be the proposition "$nm = 2(2kl)$" and u be the proposition "$nm \in \mathbb{E}$".

Theorem: If $n, m \in \mathbb{E}$ then $nm \in \mathbb{E}$.

$$(p \wedge q) \rightarrow u$$

Proof: Assume $(p \wedge q)$. *(hypothesis)*
So r and s. *(simplification, definition of $\mathbb{E}$)*
Then t. *(associative, commutative, closure of $\mathbb{N}$ under $\cdot$)*
So u. *(definition of $\mathbb{E}$)* ■

Proofs with Set of Numbers

- Can we prove this theorem about the odd numbers $\mathbb{O}$?

Theorem: If $n, m \in \mathbb{O}$ then $nm \in \mathbb{O}$.

Proof: Assume $n, m \in \mathbb{O}$.

So $n = 2k + 1$ and $m = 2l + 1$ where $k, l \in \mathbb{N}$.

Then $nm = (2k + 1)(2l + 1) = 4kl + 2k + 2l + 1 = 2(2kl + k + l) + 1$.

So $nm \in \mathbb{O}$. ■

- This proof demonstrates the closure of $\mathbb{O}$ under the operation $\cdot$.
  - It relies on the definition of $\mathbb{O}$, the associative and commutative laws of arithmetic and the closure of $\mathbb{N}$ under the operation $\cdot$.

Proofs with Set of Numbers

- Can we formalize this proof?

Let p be the proposition "$n \in \mathbb{O}$" and q be the proposition "$m \in \mathbb{O}$".
Let r be the proposition "$n = 2k + 1$" and s be the proposition "$m = 2l + 1$" where $k, l \in \mathbb{N}$.
Let t be the proposition "$nm = 2(2kl + k + l) + 1$" and u be the proposition "$nm \in \mathbb{O}$".

Theorem: If $n, m \in \mathbb{O}$ then $nm \in \mathbb{O}$.

$$(p \wedge q) \rightarrow u$$

Proof: Assume $(p \wedge q)$. *(hypothesis)*
So r and s. *(simplification, definition of $\mathbb{O}$)*
Then t. *(associative, commutative, closure of $\mathbb{N}$ under $\cdot$)*
So u. *(definition of $\mathbb{O}$)* ■

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Proof Strategies

Types of Proof

- There are several different kinds of proof.
  - We have already seen formal and informal direct proofs as well as a vacuous proof.

- Here are some other kinds of proof:

Definition: An exhaustive proof is a direct proof that considers all the cases one at a time.

Definition: A proof by contrapositive proves $\neg c \rightarrow \neg(p_1 \wedge p_2 \wedge \cdots \wedge p_n)$ instead of $(p_1 \wedge p_2 \wedge \cdots \wedge p_n) \rightarrow c$.

Definition: A proof by contradiction assumes a hypothesis p and reaches a contradiction so we conclude that $\neg p$.

- Proofs by contrapositive and contradiction are also considered indirect proofs.

Definition: A proof by induction uses mathematical induction to prove statements are true for infinite cases.

Definition: A proof by diagonalization can show that two sets of infinite size are still unequal in size.

Universal Operators

- Let $O = \{\neg, \wedge, \vee, \rightarrow, \leftarrow, \leftrightarrow, \oplus\}$ be the set of logical operators.
  - Note: We could add infinite new operators to this set.
- How do we know if there are not important operators missing?

p	q	p 😊 q
T	T	T
T	F	F
F	T	T
F	F	F

- We want to prove that some finite set of operators can capture all possible logical formulas.
  - If we do this, we say the set of operators is universal.

- Our first goal will be to show that every Boolean formula can be expressed using only operators $O_1 = \{\neg, \wedge, \vee\}$.

Disjunctive Normal Form

Theorem: If x is a Boolean formula expressed using any logical operators it can be expressed by a logically equivalent Boolean formula y using only operators from the set $O_1 = \{\neg, \wedge, \vee\}$.

Proof: Assume x is a Boolean formula expressed using any logical operator and assume $o(x_1, x_2, \ldots, x_n)$ is one such logical operator not in O_1 defined on n propositions.

The definition of $o(x_1, x_2, \ldots, x_n)$ can be stated with a truth table for o. This truth table will consist of 2^n rows, each corresponding to a different assignment of truth values to the variables $x_1, x_2, \ldots, x_n$. Each row will define the value of o given that assignment.

Now consider only the rows $r_1, r_2, \ldots, r_k$ that o evaluates to T. We can say that o will be true if it is in any of these rows $(r_1 \vee r_2 \vee \cdots \vee r_k)$. Each row can be identified by the truth assignment to $x_1, x_2, \ldots, x_n$ (i.e. $r_i = (x_1 = T) \wedge (x_2 = F) \wedge \cdots \wedge (x_k = T)$). Note that $x_i = T$ is equivalent to x_i and $x_i = F$ is equivalent to $\neg x_i$.

So then x can be expressed as $y = r_1 \vee r_2 \vee \cdots \vee r_k$ where each r_i is a Boolean formula using only $\neg$ and $\wedge$. ■

Disjunctive Normal Form

Definition: We say a Boolean formula is in disjunctive normal form when it is expressed as $c_1 \vee c_2 \vee \cdots \vee c_k$ where each c_i (called a clause) is a Boolean formula using only $\neg$ and $\wedge$.

Definition: We say a Boolean formula is in conjunctive normal form when it is expressed as $c_1 \wedge c_2 \wedge \cdots \wedge c_k$ where each c_i (called a clause) is a Boolean formula using only $\neg$ and $\vee$.

- The last theorem establishes that every Boolean formula can be expressed in disjunctive normal form. Every Boolean formula can also be expressed in conjunctive normal form.
- This also establishes that $O_1 = \{\neg, \vee, \wedge\}$ is a universal set of logical operators.

NAND

Definition: Let p and q be propositions. Then p NAND q, denoted $p \overline{\wedge} q$, is equivalent to $\neg(p \wedge q)$.

Theorem: The set $O_2 = \{\overline{\wedge}\}$ is universal.

Proof: We will show that each of the universal operations in $O_1 = \{\neg,\wedge,\vee\}$ can be written using only $\overline{\wedge}$.

Case 1: $\neg p \equiv p \overline{\wedge} p$.

p	$\neg p$	$p \wedge p$	$p \overline{\wedge} p$
T	F	T	F
F	T	F	T

Case 2: $p \vee q \equiv (p \overline{\wedge} p) \overline{\wedge} (q \overline{\wedge} q)$.

$$p \vee q \equiv \neg\neg(\mathrm{p} \vee q) \equiv \neg(\neg p \wedge \neg q) \equiv \neg p \overline{\wedge} \neg q \equiv (p \overline{\wedge} p) \overline{\wedge} (q \overline{\wedge} q)$$

Case 3: $p \wedge q \equiv (p \overline{\wedge} q) \overline{\wedge} (p \overline{\wedge} q)$.

$$p \wedge q \equiv \neg\neg(p \wedge q) \equiv \neg(p \overline{\wedge} q) \equiv (p \overline{\wedge} q) \overline{\wedge} (p \overline{\wedge} q)$$

By Lemma's 1-3 each operator in O_1 can be expressed as a formula using only operators from $O_2 = \{\overline{\wedge}\}$. Since O_1 is universal O_2 is universal. ■

Biconditional Proofs

- A proof of a biconditional statement $p \leftrightarrow q$ is usually broken into two proofs of $p \rightarrow q$ and $q \rightarrow p$.
  - When proving $S = T$ we usually prove $S \subseteq T$ and $T \subseteq S$.
  - When proving $a = b$ we usually prove $a \leq b$ and $b \leq a$.

Theorem: $3n + 2$ is odd if and only if n is odd.

Lemma: If n is odd then $3n + 2$ is odd.

Lemma: If $3n + 2$ is odd then n is odd.

Biconditional Proofs

Lemma: If n is odd then $3n + 2$ is odd.

Proof: Assume n is odd. So $n = 2k + 1$ for some $k \in \mathbb{N}$. So

$$\begin{aligned} 3n + 2 &= 3(2k + 1) + 2 \\ &= 6k + 3 + 2 \\ &= 6k + 5 \\ &= 6k + 4 + 1 \\ &= 2(3k + 2) + 1 \end{aligned}$$

where $3k + 2 \in \mathbb{N}$. So $3n + 2$ is odd. ■

Biconditional Proofs

Lemma: If $3n + 2$ is odd then n is odd.

Proof: Assume n is even. So $n = 2k$ for some $k \in \mathbb{N}$. So

$$\begin{aligned} 3n + 2 &= 3(2k) + 2 \\ &= 6k + 2 \\ &= 2(3k + 1) \end{aligned}$$

where $3k + 1 \in \mathbb{N}$. So $3n + 2$ is even. ■

- This is an example of a proof by contrapositive. Instead of proving $p \to q$ we have proven $\neg q \to \neg p$ which is logically equivalent.

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Abstract Algebra

Sets and Operators

- Recall the sets $\mathbb{E}$ and $\mathbb{O}$ the sets of even and odd natural numbers and $\cdot$ the multiplication operation.
  - We have proven that the product of an even and an even is an even, or:
$$a \in \mathbb{E} \wedge b \in \mathbb{E} \rightarrow a \cdot b \in \mathbb{E}$$
  - We have proven that the product of an odd and an odd is an odd, or:
$$a \in \mathbb{O} \wedge b \in \mathbb{O} \rightarrow a \cdot b \in \mathbb{O}$$

Definition: A set S is closed under an operation o, if and only if, if $s_1, s_2, \ldots s_n \in S$ then $o(s_1, s_2, \ldots, s_n) \in S$.

- So the even and odd numbers are closed under multiplication. Are they closed under addition?
  - The even numbers are but the odd numbers are not! Can we prove this?

Sets of Numbers

- The set of natural numbers $\mathbb{N}$ are "natural" because they are the counting numbers (a.k.a. the cardinal numbers).
  - Sometimes 0 is omitted from these numbers as we did not add a concept of zero until the 6th century!

Theorem: The set of natural numbers $\mathbb{N}$ is closed under addition and multiplication.

- We will not prove this theorem. Instead we will take it as an axiom.

Theorem: The set of natural numbers $\mathbb{N}$ is not closed under subtraction.

Proof: Assume for the sake of contradiction that $\mathbb{N}$ is closed under subtraction. Then if $a \in \mathbb{N}$ and $b \in \mathbb{N}$ then $a - b \in \mathbb{N}$. Consider $0 \in \mathbb{N}$ and $1 \in \mathbb{N}$ then $0 - 1 = -1 \notin \mathbb{N}$. This is a contradiction so $\mathbb{N}$ is not closed under subtraction. ■

- This is a proof by counter-example and an existence proof.

Inverse Operations

- Addition and subtraction are inverse operations.

$$(x + y) - y = x$$

Definition: Let $o(x)$ be an operation. The inverse operation $o^{-1}(x)$ is such that $o^{-1}\big(o(x)\big) = x$.

- What is the inverse to multiplication?

$$\frac{x \cdot y}{y} = x$$

- What is the inverse to exponentiation?

$$2^{\log_2 n} = n$$

- What is the inverse to squaring?

$$\sqrt{n^2} = n$$

  - Note: This is only true in $\mathbb{N}$. Why is it not true in $\mathbb{Z}$?

The Set of Integers

- The set of integers $\mathbb{Z}$ is called the closure of the set of natural numbers $\mathbb{N}$ under the operation of subtraction.
  - That is, if we wanted the set of natural numbers $\mathbb{N}$ to be closed under subtraction we would have to extend it to include the negative numbers $\mathbb{Z}^-$ and thus we end up with the set of integers $\mathbb{Z} = \mathbb{N} \cup \mathbb{Z}^-$.
  - Notice this means that $\mathbb{N}$ and $\mathbb{Z}^-$ partition $\mathbb{Z}$.

Definition: The additive inverse of $a \in S$, denoted $-a$, is the element $b \in S$ such that $a + b = 0$.

- Here 0 is serving as the identity element.

Definition: The identity element $0 \in S$ is such that if $a \in S$ then $a + 0 = a$.

- When the operation $\cdot$ is used instead commonly 1 is used as the identity element.

Definition: An operation $o(a, b)$ on a set S is called invertible, if and only if, for every $a \in S$ there is an element $-a \in S$ such that $o(a, -a) = 0$.

- The inverse element is usually denoted a^{-1} when we use $\cdot$ as the operation.

Groups

Definition: An algebraic group $(S, +)$ is a set S and an operation $+$ such that:

1. The set S is closed under the operation $+$.
2. The operation $+$ is associative (i.e. $(a + b) + c = a + (b + c)$).
3. There is an identity element 0 such that $a + 0 = a$.
4. The operation $+$ is invertible.

If $+$ is also commutative then $(S, +)$ is called a commutative group or an Abelian group.

- Notice that $(\mathbb{Z}, +)$ is an algebraic (Abelian) group. Is $(\mathbb{Z}, \cdot)$? Can we prove or prove not?
  - No, it is not. $\cdot$ is not invertible in $\mathbb{Z}$. But $(\mathbb{Z}, \cdot)$ satisfies the other three properties.
  - $(\mathbb{Z}, +, \cdot)$ is called a commutative ring in abstract algebra, the study of algebraic structures.
- Is $(\mathbb{N}, +)$ an algebraic group?

Modular Arithmetic

Definition: $a \bmod b$, denoted $a \,\%\, b$, is the remainder of a divided by b.

Definition: The set of integers modulo n, denoted $\mathbb{Z}_n$, is the set of remainders $\{0,1,\ldots,n-1\}$ after division by n.

- Consider $\mathbb{Z}_5 = \{0,1,2,3,4\}$. We can define the standard operations on $\mathbb{Z}_5$ if we do $\bmod 5$ after each operation.
  - What is $3 + 2$?
  - What is $3 \cdot 3$?
  - What is x if $x + 4 = 0$?
- Note that $\mathbb{Z}_n$ is closed under $+$ and $\cdot$ trivially as we do modulo n after each operation.
- Is $\mathbb{Z}_n$ a group under $+$? What about under $\cdot$?

$(\mathbb{Z}_n, +)$ is a group

Theorem: $(\mathbb{Z}_n, +)$ is an Abellian group.

Proof: We will prove that $(\mathbb{Z}_n, +)$ has each of the four properties of an algebraic group.

Case 1: Closure

Since we take modulo n after every addition we always get a new $a \in \mathbb{Z}_n$.

Case 2: Identity

0 is the identity. $a + 0 = a$ for all $a \in \mathbb{Z}_n$.

Case 3: Inverse

The inverse of 0 is 0 since $0 + 0 = 0$.

The inverse of $i \in \{1,2, \dots, n-1\}$ is $n - i$ since $i + n - i = n \equiv 0 \ \% \ n$.

Case 4: Associative/Commutative

$$(a + b) + c = a + (b + c)$$
$$a + b = b + a$$

We assume addition is associative and commutative as axioms.

So $(\mathbb{Z}_n, +)$ is an Abellian group. ■

$(\mathbb{Z}_n,\cdot)$ is not a group

Theorem: $(\mathbb{Z}_n,\cdot)$ is not a group.

Proof: We will prove that $(\mathbb{Z}_n,\cdot)$ has each of the four properties of an algebraic group *except* inverse.

Case 1: Closure

Since we take modulo n after every multiplication we always get a new $a \in \mathbb{Z}_n$.

Case 2: Identity

1 is the identity. $a \cdot 1 = a$ for all $a \in \mathbb{Z}_n$.

Case 3: Inverse

0 has no inverse. $0 \cdot a = 0$ for all $a \in \mathbb{Z}_n$ so there is no inverse to 0.

Case 4: Associative/Commutative

$$(a \cdot b) \cdot c = a \cdot (b \cdot c)$$
$$a \cdot b = b \cdot a$$

We assume multiplication is associative and commutative as axioms.

So $(\mathbb{Z}_n,\cdot)$ satisfies all criteria of a group but inverse. ■

The Set of Rational Numbers

- The set of rational numbers $\mathbb{Q}$ is the closure of the set of integers $\mathbb{Z}$ under the operation of division.

Definition: The set of rational numbers is $\mathbb{Q} = \left\{\frac{p}{q} \mid p \in \mathbb{Z} \wedge q \in \mathbb{Z}^+\right\}$.

- Can we prove that $\mathbb{Z} \subset \mathbb{Q}$?

Theorem: $\mathbb{Z} \subset \mathbb{Q}$

Proof: We must prove $\mathbb{Z} \subseteq \mathbb{Q}$ and $\mathbb{Z} \neq \mathbb{Q}$.

Lemma 1: $\mathbb{Z} \subseteq \mathbb{Q}$.

To prove this we must prove the statement if $p \in \mathbb{Z}$ then $p \in \mathbb{Q}$. Assume $p \in \mathbb{Z}$. Then $p = \frac{p}{q}$ when $q = 1 \in \mathbb{Z}^+$. So $\mathbb{Z} \subseteq \mathbb{Q}$.

Lemma 2: $\mathbb{Z} \neq \mathbb{Q}$.

To prove this we must find any $p \in \mathbb{Q}$ such that $p \notin \mathbb{Q}$. For instance $\frac{1}{2} \in \mathbb{Q}$ and $\frac{1}{2} \notin \mathbb{Z}$. So $\mathbb{Z} \neq \mathbb{Q}$.

So by Lemma 1 and Lemma 2 $\mathbb{Z} \subset \mathbb{Q}$. ■

- $\mathbb{Q}$ is closed under addition, subtraction, multiplication and division (you prove this on Assignment 2).
- In abstract algebra a set that has addition, subtraction, multiplication and division – all operating normally – is called a field.
- $\mathbb{Q}$, $\mathbb{R}$, and $\mathbb{C}$ are all fields but $\mathbb{N}$ and $\mathbb{Z}$ are not.

Proof by Contradiction

- In a proof by contradiction we assume the negation of what we are trying to proof $\neg p$ in order to reach a contradiction. The result is that what we are trying to prove p must be the case.

Theorem: $\sqrt{2}$ is irrational.

Proof: Assume for the sake of contradiction that $\sqrt{2}$ is rational.

Then $\sqrt{2} = \frac{p}{q}$ for $p, q \in \mathbb{N}$ where $q \neq 0$ and where p and q share no common factors (i.e. $\frac{p}{q}$ is in lowest terms).

If we square both sides we get that $2 = \frac{p^2}{q^2}$ or that $2q^2 = p^2$. Thus p^2 is even and so p is even.

If p is even then $p = 2k$ and $p^2 = 4k^2$. So $2q^2 = 4k^2$ or $q^2 = 2k^2$. So q is even.

If both p and q are even then they share a common factor which is a contradiction to our assumption that the do not share a common factor. Therefore our initial assumption must be false and $\sqrt{2}$ must be irrational. ■

(Ir)Rational Numbers and Decimal Expansion

- The set of real numbers $\mathbb{R}$ includes the rational numbers $\mathbb{Q}$ as well as the irrational numbers.
  - One characteristic of irrational numbers is when written using decimal notation they require infinite non-repeating numerals.
    - $\sqrt{2} = 1.4142135623\ldots$
    - $\pi = 3.1415926535\ldots$
    - $e = 2.7182818284\ldots$
    - $\phi = 1.6180339887\ldots$
  - Rational numbers, in contrast, can be written with only finite non-zero numerals, or with infinite repeating numerals.
    - $\frac{1}{2} = 0.5$
    - $\frac{1}{3} = 0.\overline{3}$
- Is there any way we can write $\frac{1}{3}$ with only a finite number of numerals?

(Ir)Rational Numbers and Decimal Expansion

- Is there any way we can write $\frac{1}{3}$ with only a finite number of numerals?
  - Not in base 10. But yes if we change base. What base should we change to?
- Recall that we write numbers in base 10 for convenience, but that they can be written in different bases like binary, octal or hexadecimal.
  - What is 100 is binary?

$$100_{10} = 1100100_2$$

  - Quickly convert this binary number to octal and hex.

$$100_{10} = 1100100_2 = 144_8 = 64_{16}$$

- We can choose any positive natural number b as our base to write a number down.
  - Write 100 in base 1.

$$100_{10} = \overbrace{000\cdots0}^{100}$$

  - Write 100 in base 3.

$$100_{10} = 10201_3$$

- While we typically use other bases to express integer values, we can also use them with the decimal notation to express rational numbers.
  - Write $\frac{1}{3}$ in base 3.

$$\frac{1}{3} = 0.1_3$$

(Ir)Rational Numbers and Decimal Expansion

Theorem: If $n \in \mathbb{Q}$ then n can be expressed using the decimal system using finite non-zero numerals in some base.

Proof: Assume $n \in \mathbb{Q}$. Then $n = \frac{p}{q}$ where $p, q \in \mathbb{Z}$ and $q > 0$.

Assume without loss of generality that $q > 1$ and $|p| < q$ (we know integers and the integer component of a rational number can be written with finite non-zero numerals in any base).

Now consider writing n in base q. Since we are in base q the first numeral after the decimal is the $\frac{1}{q}$ place and so $n = \frac{p}{q} = 0.p_q$.

Thus n can be written with finite non-zero numerals in base q. ■

Beyond the Rational Numbers

- The set of real numbers $\mathbb{R}$ includes the rational numbers $\mathbb{Q}$ as well as the irrational numbers.
  - The irrational numbers are usually just defined as $\mathbb{R} \setminus \mathbb{Q}$.

- The set of complex numbers $\mathbb{C}$ are the closure of the real numbers under the operation of square root.
  - $i = \sqrt{-1}$ is an imaginary number. All complex numbers can be expressed as $\alpha + \beta i$ where $\alpha, \beta \in \mathbb{R}$.

- Other sets of numbers exist beyond the complex numbers.
  - Quaternions
  - Octonians